

Absolutely! Let's make a **detailed, step-by-step lab guide** for your BIS practical exam in **Power BI**, based on the tasks you shared. I'll organize it clearly so you can follow it in the lab without confusion.

BIS Practical Lab Guide – Power BI Dashboard

Task 1: Data Loading and Modeling

Step 1: Load the Datasets

1. Open **Power BI Desktop**.
 2. Click **Home** → **Get Data** → **Excel** → **Connect**.
 3. Select the files:
 - Dashboard Order.xlsx
 - Dashboard Details.xlsx
 4. In **Navigators**, check the tables you want:
 - Table 1: Details
 - Table 2: Merge1
 - Table 3: Order
 5. Click **Load**.
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Step 2: Merge Data (Join Tables)

1. Go to **Home** → **Transform Data** → **Power Query Editor**.
 2. Select `Details` table.
 3. Click **Merge Queries** → **Merge Queries as New**.
 4. Select `Order` table to merge with.
 5. Match **Order ID** column in both tables.
 6. Click **OK** → This creates a merged table (you can rename it `MergedTable`).
 7. Close & Apply changes.
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Step 3: Create Date Columns

1. Go to **Data View** → Select '**Order**' table.
2. Create new calculated columns:

Month Name

```
Month Name = FORMAT('Order'[Order Date], "MMMM")
```

Day of Week

```
Day of Week = FORMAT('Order'[Order Date], "dddd")
```

Optional for sorting weekdays (Mon → Sun):

Day Number

```
Day Number = WEEKDAY('Order'[Order Date], 2)
```

- 2 means Monday = 1, Sunday = 7 (useful for sorting charts by weekday).
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Task 2: Create Logic-Based Metrics (Measures)

Step 1: Weekend Sales

1. Go to **Modeling** → **New Measure**.
2. Enter:

```
Weekend Sales =  
CALCULATE(  
    SUM(Details[Amount]),  
    'Order'[Day of Week] IN {"Saturday", "Sunday"}  
)
```

- This sums all sales on **Saturday** and **Sunday**.
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Step 2: Profit Ratio

1. **New Measure**:

```
Profit Ratio = DIVIDE(SUM(Details[Profit]), SUM(Details[Amount]), 0)
```

- Format as **Percentage** in the Measure tools.
 - Display on **KPI card** later.
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Step 3: Average Order Value (AOV)

1. New Measure:

`Avg Order Value = DIVIDE(SUM(Details[Amount]), COUNT(Details[Order ID]), 0)`

- This gives average value per order.
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Task 3: Create Visualizations

1 Sales Trends by Weekday (Column Chart)

1. Insert → **Clustered Column Chart**.
 2. **Axis:** 'Order'[Day of Week]
 3. **Values:** SUM(Details[Amount])
 4. Sort weekdays properly:
 - Click 'Order'[Day of Week] → **Column tools** → Sort by 'Order'[Day Number]
 5. Optional: Enable **Data Labels**.
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2 Monthly Sales Performance (Line Chart)

1. Insert → **Line Chart**.
 2. **X-Axis:** 'Order'[Month Name]
 3. **Values:** SUM(Details[Amount])
 4. Sort months chronologically:
 - Select 'Order'[Month Name] → **Column tools** → Sort by 'Order'[Month Number]
 5. Optional: Add **Data Labels & Title** → "Monthly Sales Performance"
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3 Top 5 Customers by Profit (Bar Chart)

1. Insert → **Clustered Bar Chart**.
2. **Axis:** 'Order'[CustomerName]
3. **Values:** SUM(Details[Profit])
4. Filter Top 5:
 - **Filters pane** → CustomerName → Top N → Show Top 5 by **Sum of Profit** → Apply
5. Sort descending → highest profit on top

4 Payment Mode Preference (Pie Chart)

1. Insert → **Pie Chart**.
2. **Legend:** `Details[PaymentMode]`
3. **Values:** `COUNT(Details[Order ID])` (or `SUM(Amount)` for revenue)
4. Add **Data Labels & Title** → "Payment Mode Preference"

Task 4: Add Interactive Features (Slicers)

1 Category Slicer

1. Insert → **Slicer**
2. **Field:** `Details[Category]`
3. Resize on dashboard
4. Test: Click a category → All visuals update automatically
5. Optional: Change slicer type → **Dropdown** (for cleaner look)

2 City Filter (Dropdown)

1. Insert → **Slicer**
 2. **Field:** `'Order'[City]`
 3. Change slicer type → **Dropdown**
 4. Resize on dashboard
 5. Test: Select a city → All visuals update
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